Association of dietary soy genistein intake with lung function and asthma control: a post-hoc analysis of patients enrolled in a prospective multicentre clinical trial

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Background Broad dietary patterns have been linked to asthma but the relative contribution of specific nutrients is unclear. Soy genistein has important anti-inflammatory and other biological effects that might be beneficial in asthma. A positive association was previously reported between soy genistein intake and lung function but not with asthma exacerbations. Aims To conduct a post-hoc analysis of patients with inadequately controlled asthma enrolled in a prospective multicentre clinical trial to replicate this association. Methods A total of 300 study participants were included in the analysis. Dietary soy genistein intake was measured using the Block Soy Foods Screener. The level of soy genistein intake (little or no intake, moderate intake, or high intake) was compared with baseline lung function (pre-bronchodilator forced expiratory volume in 1 second (FEV1)) and asthma control (proportion of participants with an episode of poor asthma control (EPAC) and annualised rates of EPACs over a 6-month follow-up period. Results Participants with little or no genistein intake had a lower baseline FEV1 than those with a moderate or high intake (2.26L vs. 2.53L and 2.47L, respectively; $p=0.01$). EPACs were more common among those with no genistein intake than in those with a moderate or high intake (54% vs. 35% vs. 40%, respectively; $p<0.001$). These findings remained significant after adjustment for patient demographics and body mass index. Conclusions In patients with asthma, consumption of a diet with moderate to high amounts of soy genistein is associated with better lung function and better asthma control.